

RIO DE JANEIRO SANTOS DUMONT, Brazil

WMO: 837550

Lat: 22.910S

Lon: 43.163W

Elev: 11

StdP: 14.69

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 62.2	(c) 62.9	(d) 53.3	(e) 60.5	(f) 67.9	(g) 55.3	(h) 65.2	(i) 68.1	(j) 19.7	(k) 73.0	(l) 17.2	(m) 73.5	(n) 6.2	(o) 310	(p) 0.303

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 2	(b) 12.3	(c) 94.6	(d) 78.3	(e) 91.6	(f) 77.9	(g) 89.7	(h) 77.4	(i) 80.9	(j) 89.1	(k) 80.0	(l) 87.7	(m) 79.2	(n) 86.6	(o) 8.9	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	150.0	86.5	77.4	142.5	84.4	77.1	140.9	83.9	44.7	89.3	43.6	88.2	42.8	86.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.7	(b) 16.5	(c) 14.7	(e) 59.2	(f) 99.9	(g) 2.2	(h) 1.4	(i) 57.6	(j) 100.9	(k) 56.3	(l) 101.7	(m) 55.1	(n) 102.5	(o) 53.5	(p) 103.5		
			DB	WB	54.9	82.0	2.7	1.8	53.0	83.3	51.4	84.4	49.9	85.4	48.0	86.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.6	81.4	82.1	80.6	78.6	74.4	72.8	71.9	72.5	73.4	75.4	76.8	79.5
	DBStd	5.27	4.29	3.51	3.27	3.52	3.49	3.61	4.01	3.96	4.22	4.12	4.51	4.05
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	5	0	0	0	0	0	0	3	1	1	0	0	0
	CDD50	9704	972	900	950	857	756	684	679	697	702	787	804	916
	CDD65	4235	507	480	485	407	291	235	217	233	253	322	354	451
	CDH74	30739	5114	5045	4519	3124	1278	799	723	889	1127	1797	2424	3900
	CDH80	9170	1879	1903	1397	770	203	118	118	171	251	413	688	1259
(14) Wind	WSAvg	6.8	7.2	7.2	6.6	6.3	6.1	6.1	6.3	6.7	7.3	7.6	7.4	7.3
(15) Precipitation	PrecAvg	45.4	5.4	5.7	5.3	4.5	3.2	2.0	2.0	2.0	2.6	3.4	3.9	5.8
	PrecMax	68.9	18.6	18.6	13.1	11.8	10.4	6.1	6.6	5.0	8.8	7.8	7.7	12.4
	PrecMin	23.0	1.2	0.0	1.1	0.7	0.4	0.1	0.1	0.0	0.0	0.1	1.3	2.4
	PrecStd	10.9	3.6	4.6	3.6	2.6	2.2	1.7	1.5	1.2	1.9	1.8	1.7	2.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.9	96.6	94.6	92.8	88.0	86.3	86.4	89.1	91.0	91.6	94.8	96.8
		MCWB	79.1	79.7	77.6	77.4	74.5	71.9	70.8	71.2	73.3	75.2	76.9	79.3
	2%	DB	93.3	93.1	90.0	88.2	84.4	82.8	82.8	84.3	85.9	87.4	89.9	91.6
		MCWB	79.6	78.6	78.5	77.2	74.0	71.5	70.4	70.6	72.3	75.2	76.2	77.8
	5%	DB	91.0	90.0	88.0	86.1	82.2	80.7	80.5	80.9	82.5	84.2	86.4	89.2
		MCWB	78.8	78.2	78.3	76.6	73.2	71.2	70.3	70.3	72.0	74.2	75.8	77.3
	10%	DB	88.0	88.1	86.2	84.2	80.2	78.6	77.4	78.7	80.2	82.0	84.0	86.2
		MCWB	77.9	78.1	77.4	75.6	72.6	70.9	69.6	69.7	71.3	73.5	74.8	76.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.2	82.4	81.9	80.1	78.0	75.9	75.4	74.2	76.4	78.8	79.6	81.0
		MCDB	92.1	92.0	88.6	86.9	82.7	80.9	79.5	82.3	84.1	85.7	87.9	90.4
	2%	WB	80.7	80.6	80.3	78.7	75.8	73.7	72.7	72.5	74.5	76.5	78.0	79.6
		MCDB	89.8	88.6	86.8	85.1	81.1	79.2	78.1	79.5	81.0	82.8	85.7	88.0
	5%	WB	79.6	79.7	79.3	77.5	74.4	72.4	71.5	71.5	73.0	75.3	76.8	78.3
		MCDB	87.9	87.4	85.4	83.5	79.7	77.9	77.5	78.2	79.2	81.6	83.8	86.2
	10%	WB	78.5	78.9	78.3	76.5	73.3	71.5	70.3	70.4	72.0	74.3	75.6	77.3
		MCDB	86.0	86.3	84.3	82.3	78.5	76.8	75.8	76.5	78.1	80.1	81.9	84.3
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.3	11.1	10.9	10.4	11.0	11.3	11.3	11.1	10.5	11.0	11.6
		MCDBR	16.0	15.3	13.9	13.7	13.1	14.3	15.4	15.5	16.4	15.0	16.1	16.6
		MCWBR	6.5	6.2	5.9	5.9	5.9	6.2	6.6	6.3	6.6	6.5	6.4	6.5
	5% WB	MCDBR	14.1	13.5	12.4	12.0	11.9	12.5	13.2	13.2	13.6	12.9	14.1	14.4
		MCWBR	6.6	6.5	6.2	5.7	6.0	6.2	6.4	6.1	6.3	6.1	6.2	6.5
(40) Clear-Sky Solar Irradiance	taub		0.413	0.409	0.409	0.397	0.382	0.381	0.383	0.414	0.492	0.467	0.433	0.424
	taud		2.479	2.472	2.482	2.502	2.516	2.508	2.485	2.376	2.129	2.263	2.392	2.433
	Ebn at Noon		296	293	285	274	265	259	262	265	256	273	288	293
	Edh at Noon		37	37	35	32	29	28	30	36	49	45	41	39
(44) All-Sky Solar Radiation	RadAvg		1920	1991	1700	1490	1181	1104	1144	1355	1460	1640	1693	1872
	RadStd		278	204	139	105	63	80	98	103	185	190	139	167

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	-3.04	N/A	-2.52	-2.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page